



Figure 1: Prometheus' architecture

Installing Prometheus and node exporter

We will now install Prometheus and one of the exporters called the node exporter. The node exporter is an application that runs on a node and can collect various metrics like memory, disk I/O and more. It also exposes an endpoint, which the Prometheus server scrapes at regular intervals and collects the metrics.

Visit the Prometheus downloads page at <https://prometheus.io/download/> and you will see the binaries made available for Prometheus and various other modules like the alerts manager, node exporter and more.

Assuming that you want to install the Prometheus server on a Linux distribution, download the Prometheus server as shown below:

```
wget "https://github.com/prometheus/prometheus/releases/download/v1.5.0/prometheus-1.5.0.linux-amd64.tar.gz"
```

Extract the files into a folder and, at the root of that folder, you should see the following key files: *prometheus* and *prometheus.yml*.

Now, let us go and download the node exporter as shown below:

```
wget "https://github.com/prometheus/node_exporter/releases/download/v0.13.0/node_exporter-0.13.0.darwin-amd64.tar.gz"
```

Extract the files into a folder and, at the root of that folder, you should see the following key file: *node_exporter*.

Monitoring your node with Prometheus

In the earlier section on architecture, we saw that the Prometheus server can collect metrics from multiple nodes that need to be monitored. In our example here, we will monitor the Prometheus server itself. We will execute these programs on the same node, for the sake of simplicity.

The first thing that we shall run is the *node_exporter*.

The node exporter collects multiple metrics about the node like memory, disk I/O, processes and more. To run the node exporter, go ahead and run the *node_exporter* program that you just downloaded. The output is shown below:

```
$ ./node_exporter &
[1] 484
romin_irani@promserver:~/Prometheus/node_exporter/node_exporter-0.13.0.linux-amd64$ INFO[0000] Starting node_exporter (version=
0.13.0, branch=master, revision=006d1c7922b765f458fe9b92ce646
641bded0f52) source=node_exporter.go:135
INFO[0000] Build context (go=go1.7.3, user=root@75db7098576a,
date=20161126-13:11:09) source=node_exporter.go:136
INFO[0000] No directory specified, see --collector.textfile.
directory source=textfile.go:57
INFO[0000] Enabled collectors:
source=node_exporter.go:155
INFO[0000] - mdadm
source=node_exporter.go:157
INFO[0000] - meminfo
source=node_exporter.go:157
INFO[0000] - vmstat
source=node_exporter.go:157
INFO[0000] - loadavg
source=node_exporter.go:157
INFO[0000] - entropy
source=node_exporter.go:157
INFO[0000] - filefd
source=node_exporter.go:157
INFO[0000] - netdev
source=node_exporter.go:157
INFO[0000] - sockstat
source=node_exporter.go:157
INFO[0000] - textfile
source=node_exporter.go:157
INFO[0000] - diskstats
source=node_exporter.go:157
INFO[0000] - netstat
source=node_exporter.go:157
INFO[0000] - filesystem
source=node_exporter.go:157
INFO[0000] - hmon
source=node_exporter.go:157
INFO[0000] - stat
source=node_exporter.go:157
INFO[0000] - time
source=node_exporter.go:157
INFO[0000] - uname
source=node_exporter.go:157
INFO[0000] - conntrack
source=node_exporter.go:157
INFO[0000] Listening on :9100
source=node_exporter.go:176
```